



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
(AEROSPACE ENGINEERING)

EAR 302/EMM 308: COMPUTER PROGRAMMING II

3RD TRIMESTER 2022/2023

TIME: 2 HOURS

By Mr. Patrick Njuguna.

INSTRUCTIONS:

- (i) This paper consists of **FIVE** Questions.
- (ii) Question **ONE** is Compulsory and contains 30 Marks.
- (iii) Attempt any **TWO** of the remaining **FOUR** Questions. Each Question is 20 Marks

QUESTION ONE

- (a) Briefly discuss any three object-oriented programming concepts. (6 marks)
- (b) Discuss any three limitations of components-based software development. (6 marks)
- (c) Briefly explain the architecture of an application program. (6 marks)
- (d) Design a form for capturing project details typed on textboxes and displayed on tabular format using data grid view control. Write instructions to post data to data grid view, sort them and also send to external file named projects.txt. (6 marks)
- (e) Discuss any three principles of designing graphical user interface. (6 marks)

QUESTION TWO

- (a) Discuss role of application programming interface (API) (4 marks)
- (b) Discuss any three commonly used data structures and state application of each. (6 marks)
- (c) Design a form for recording employees' details and submitting data to a database. Write a program to display all employees' Details in a data grid view control. (10 marks)

QUESTION THREE

- (a) Briefly explain structure of C# program. (4 marks)
- (b) Briefly explain why layered software tends to execute slowly. (6 marks)
- (c) Discuss any two roles of object browser in visual studio IDE. (4 marks)
- (d) Design a graphical user interface form to capture minimum and maximum radius of metal rods as well as desired rod height. GUI should be Data grid view to list various radius from minimum to maximum and corresponding volume. Write a code to display all radius between minimum and maximum and volume on data grid view. (6 marks)

QUESTION FOUR

- (a) Explain any two ways input output devices transfer data to and from CPU (6 marks)
- (b) Outline role of the following in programming:
 - (i) Device driver
 - (ii) Device driver application programming interface
 - (iii) Device driver file
 - (iv) Software APIs (8 marks)
- (c) The drag force on a car is given by

$$F_D = \frac{1}{2} \rho v^2 A C_D$$

Where ρ is the density of air (1.23 kg/m^3), v is the velocity in units of m/s, A is the projected area of the car (2.5m) and C is the drag coefficient (0.2).

Design a form ^{and write the code} for displaying drag force when used enter various values of velocity and user can also change drag coefficient based on vehicle design. (6 marks)

QUESTION FIVE

- (a) Outline how user program interacts with databases. Explain any two software components used. (6 marks)
- (b) Explain how data on window form can be transferred to database. State classes involved. Write program that capture data entered on window form and post data to SQL server database. (8 marks)
- (c) Briefly explain the difference between structure and class in program and state shortcoming of each. (6 marks)